**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 087 · Lesson 087 · Algorithm Design and Testing Challenge

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is an invariant?

Q2. Why is sorting unnecessary for just finding one maximum?

Q3. What is nullable int?

Q4. Binary search for 8 in [2,4,6,8,10]. Give midpoint indexes and result.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

maximum — Largest value in set.

invariant — Maintained fact during loop.

nullable — Type allowing absence/null.

complexity — Resource growth as input increases.

## New learning

Observe — Specify output when inputs are empty, and preserve correctness on all-negative arrays.

Explain — Maintain invariant: best is the largest value among those inspected so far.

Evaluate — Compare against alternative sorting-based solution: an O(n) scan avoids O(n log n) sort when only the maximum is needed.

## Worked example

Trace Maximum([7,3,10,4]).

1. Model response — The invariant tracks largest seen, not last examined.

# We Do · Practise with guidance

Trace different Maximum([-9,-4,-11]).

Model response

Starting best at0 would be incorrect for all-negative arrays.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Trace Maximum([-5,-2,-8]) showing all changes and final value. [6]

### Question 2

Write algorithm for maximum of array, including empty input case. [7]

### Question 3

Compare O(n) maximum scan with sorting first and describe useful test coverage. [6]

# Exit Ticket · Answer before checking

Initialising maximum to zero works for every integer array.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. A condition maintained while processing data.

Q2. A single scan can find it in O(n).

Q3. An int? that can be an integer or null.

Q4. mid2=6, then mid3=8; found index3.

### You Do pages · Independent questions

Q1. Initial−5 [2], becomes−2 [2], remains−2 [1], returns−2 [1].

Q2. Empty return null [2], initial best=a[0] [1], iterate array [1], update if x>best [2], return best [1].

Q3. Scan O(n) [2], sorting typically O(n log n) [2], test empty and all-negative cases [2].

### Exit Ticket · Model answer

False. It fails on arrays containing only negative values.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**